

1(a) A list of all implemented classes with a brief description what the purpose / function of that class is

- CANFrame: Holds the CAN data (i.e.. Time Offset, Message Id, Data Length and Data Bytes)
- CANSimulation: It orchestrates the parsing of CAN data File and GPS coordinates file and it also take cares for starting up HTTP server and Websocket for GUI simulation
- CANTrace: It holds the parsed CAN data and provides functionality to access CAN data on need.
- CANTraceParser: Is used to parse CAN data from the file given by the user
- GPSCoordinate: Is used to store GPS coordinates
- GPSParser: Is used to parse GPS data from the file given by the user
- GPSTrace: It holds the parsed GPS data and provides functionality to access GPS data based on time offset.
- HttpServerHandler: Handles the requests for the root endpoint by returning the index.html file
- MultipleCANFrameData: Is blueprint for storing multiple sensor data in a single CAN frame
- SensorDataReceiver: Receives the sensor data from CANSimulation.startSimplulation and transfers it to the websockets for broadcasting of data
- SimulationGUI: Is responsible for starting up HTTP server and Websocket
- SingleCANFrameData: Is blueprint for storing single sensor data in a CAN frame
- SocketHandler: Process the requests coming through the sockets (i.e.. Making new connection, Broadcasting data)

(b) A list of all group members and for each member what they worked on (e.g. which functions, classes, and similar the member worked on)

- Keerthana:
 - Index.html
 - HttpServerHandler
 - SensorDataReceiver
 - GPSParser
 - CANSimulation
- Manoj:
 - GPSCoordinate
 - GPSTrace
 - SocketHandler
 - SimulationGUI
 - SocketHandler
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(c) A screenshot of the console output (should show proper sensor values at any point during the simulation)

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PS C:\Users\Manoj A M\Desktop\Assignment\SEM 1\Internet Embedded System\Assignment 1\New> & 'C:\Users\Manoj A M\.vscode\extensions\redhat.java-1.29.0-win32-x64\17.0.10-win32-x86_64\bin\java.exe' '@C:\Users\MANOJA~1\AppData\Local\Temp\cp_75b45ns4pjcxm2077992p3yrb.argfile' 'assignment.CANSimulation'
website running on: http://127.0.0.1:56161/
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
Please enter the following port number inside the website when prompted: 56162
WebSocket server started successfully
Received message: START SIM from /[0:0:0:0:0:0:1]:56197
| Current Time | Vehicle Speed | SteerAngle | YawRate | LatAccel | LongAccel | GPS Lat/Long |
| 3067.4       | 74.9          | ...        | ...     | ...      | ...       | 52.721302 / 13.22434 ||
```

(d) A screenshot of the GUI while simulating (should show proper sensor values at any point during the simulation)

